

HelixCode Architecture

Scope: Factual, verified architecture map of the HelixCode repository. This document is the authority referenced by CLAUDE.md §11 for architecture questions. It describes what EXISTS in the live tree (verified by inspecting .gitmodules, both go.mod files, and directory listings); it is a structural map, not a status report. Known-incomplete areas are flagged honestly in the final section.

Note on CLAUDE.md drift: CLAUDE.md §3.2 describes an older, partly flat submodule layout (e.g. challenges/, containers/, security/, helix_qa/, helix_agent/ at repo root, and a dependencies/ group for the HelixDevelopment modules). The live tree has moved nearly all owned submodules under submodules/<name>/. Where §3.2 and the live tree disagree, this document follows the live tree and notes the discrepancy inline.

1. Repository topology

HelixCode is a **governance / meta-repository** that contains a thin root Go module, a tracked inner Go application, and a large set of Git submodules.

1.1 Two Go modules — both module dev.helix.code

Layer	Path	module	go version	Role
Root (thin)	go.mod	dev.helix.code	1.25.2	Governance/ meta module. Only 3 require lines (google/ uuid, pkg/ errors, yaml.v2), all // indirect. Hosts root-level helpers (internal/, cmd/) and governance gates.
Inner (full app)	helix_code/ go.mod	dev.helix.code	1.26	The real HelixCode Go application: server, CLI,

Layer	Path	module	go version	Role
				internal domain packages, applications, tests, and all transitive third-party + own-org dependencies.

Both modules declare the same module path `dev.helix.code`. They are distinct modules living one directory apart.

1.2 `helix_code/` is a tracked subdirectory, NOT a submodule

Verified: `.gitmodules` contains **zero** entries for `helix_code` (`grep -c helix_code .gitmodules` → 0). The inner application is a normal tracked subdirectory of the meta-repo, consistent with CLAUDE.md §3.2.1's "circular reference if promoted" note. Instructions that reference `internal/auth`, `cmd/cli`, etc. almost always mean `helix_code/internal/...` / `helix_code/cmd/...`.

1.3 Submodule layout — `submodules/<name>/` (live tree), differs from CLAUDE.md §3.2

`.gitmodules` defines submodules across several path families:

- **`submodules/<name>/`** — the grouped layout for owned modules. 63 directories present under `submodules/`. NOTE the `.gitmodules` *section names* are inconsistent with the *paths*: e.g. the section `[submodule "containers"]` has `path = submodules/containers`, and `[submodule "dependencies/HelixDevelopment/llm_orchestrator"]` has `path = submodules/llm_orchestrator`. The **path** is authoritative; the live checkout puts these under `submodules/`. This is the principal divergence from CLAUDE.md §3.2, which still lists `challenges/`, `containers/`, `security/`, `helix_qa/`, `helix_agent/` as flat repo-root entries.
- **`dependencies/<name>/`** — third-party upstream mirrors: `dependencies/LLama_CPP` (ggml-org/llama.cpp), `dependencies/Ollama` (ollama/ollama), `dependencies/HuggingFace_Hub` (huggingface/huggingface_hub).
- **`cli_agents/<name>/`** — 50 third-party reference CLI-agent repos (aider, crush, gemini-cli, qwen-code, plandex, open-interpreter, swe-agent, gpt-engineer, claude-code, etc.).
- **`cli_agents_resources/<name>/`** — reference resource repos (Awesome-AI-Agents, Awesome-AI-GPTs, OpenAI-Cookbook, Cheshire-Cat-Ai docs, GitHub-Awesome-Copilot, Taches-CC-Resources).
- **Repo-root single submodules** — `github_pages_website` (HelixDevelopment-Code/Welcome marketing site) and `awesome-ai-memory` (topoteretes/awesome-ai-memory).

Only a subset of the 63 `submodules/*` directories are wired as formal `.gitmodules` entries; the remainder are present in the working tree (see §7 for the honest gap on this).

1.4 Inner module wiring via replace directives

helix_code/go.mod consumes own-org submodules through local replace directives pointing at ../submodules/<name> — e.g.:

```
replace digital.vasic.containers      => ../submodules/containers
replace digital.vasic.helixqa         => ../submodules/helix_qa
replace digital.vasic.docprocessor    => ../submodules/
doc_processor
replace digital.vasic.llmorchestrator => ../submodules/
llm_orchestrator
replace digital.vasic.visionengine    => ../submodules/
vision_engine
replace digital.vasic.challenges      => ../submodules/challenges
replace digital.vasic.security        => ../submodules/security
replace digital.vasic.debate          => ../submodules/
debate_orchestrator
replace digital.vasic.helixspecifier  => ../submodules/
helix_specifier
replace digital.vasic.lazy            => ../submodules/lazy
```

The inner require block lists own-org modules digital.vasic.containers, digital.vasic.debate, digital.vasic.helixqa, digital.vasic.helixspecifier, digital.vasic.lazy (versions v0.0.0-... resolved via the local replace).

2. Inner application structure (helix_code/)

2.1 helix_code/cmd/* — entry points and command surfaces

Verified contents (mix of subdirectories and root-level .go files):

- cli/ — CLI client entry.
- server/ — HTTP server entry.
- helix_config/, config_test/ — config tool and config validator.
- security_fix/, security_fix_standalone/, security_scan/, security_test/ — security tooling.
- performance_optimization/ — performance tools.
- infrastructure/, i18n/ — infrastructure and i18n command surfaces.
- Root-level .go files: root.go, main_commands.go, other_commands.go, local_llm.go, local_llm_advanced.go, i18n_seam.go, plus *_test.go files (e.g. local_llm_i18n_test.go, cmd_residual_i18n_round453_test.go).

2.2 helix_code/internal/* — domain packages (74 verified)

Each is the real domain implementation (the internal/mocks/ package is unit-test-only):

adapters	agent	approval	approvalwire	auth
autocommit	cache	clarification	cognee	commands
config	context	continua	database	deployment
discovery	editor	event	fix	focus
hardware	helixqa	hooks	i18n_wiring	i18nwiring
kilocode	llm	logging	logo	mcp
memory	mocks	monitoring	notification	performance
persistence	planner	plantree	plugins	pprofutil
project	projectmemory	provider	providers	quality
redis	render	repomap	roocode	rules
secrets	security	server	session	task
telemetry	template	testutil	theme	tools
verifier	version	voice	worker	workflow
workspace				

Role groupings: - **LLM / providers:** llm, provider, providers, verifier (LLMsVerifier integration, CONST-036–040), cognee. - **Agent / orchestration:** agent, commands, task, worker, workflow, planner, plantree, continua, self-improvement-adjacent quality. - **Context / code intel:** context, repomap, discovery, project, projectmemory, memory, editor. - **IO / integration surfaces:** mcp, hooks, plugins, tools, voice, render. - **Platform / infra:** database, redis, persistence, cache, server, deployment, hardware, monitoring, telemetry, performance, pprofutil, logging, event. - **Security / auth:** auth, security, secrets, approval, approvalwire. - **i18n / theming / branding:** i18n_wiring, i18nwiring, theme, logo. - **Adapter shims for ported agents:** kilocode, roocode, adapters, focus, clarification, autocommit. - **Test-only:** mocks, testutil.

2.3 helix_code/applications/* — platform front-ends

android, aurora_os, desktop (Fyne GUI), harmony_os, ios, terminal_ui (tview/tcell TUI), plus a README.md. NOTE the directory names use snake_case (aurora_os, harmony_os, terminal_ui) per CONST-052, differing from CLAUDE.md §3.2.1’s hyphenated terminal-ui / aurora-os / harmony-os listing.

2.4 helix_code/tests/* — test layers

unit (mocks allowed here only), integration (–tags=integration), e2e, automation, regression, security, performance, stresschaos, memory, qa, infrastructure, testinfra, plus COVERAGE_REPORT.md, TEST_COVERAGE_REPORT.md, TESTING_INFRASTRUCTURE.md, README.md.

2.5 Other inner directories

api/, pkg/, shared/, config/, docker/, scripts/, security/, assets/, examples/, benchmarks/, packaging/. Several docker-compose.*.yaml files exist at the inner root, including docker-compose.full-test.yaml (the zero-skip integration stack referenced by the test targets).

3. Submodule catalogue (submodules/*)

The grouped owned-module layout. Roles inferred from each submodule's own README.md first heading and directory name (verified live). Modules whose README opens with a `digital.vasic.* / Helix*` identifier are owned-org reusable libraries.

Submodule	Role (one-line)
helix_qa	QA / Challenge-orchestration platform (HelixDevelopment/HelixQA).
challenges	Cross-cutting Challenge bank (vasic-digital/Challenges).
containers	Container/runtime abstraction (Docker/Podman/Qemu) — vasic-digital/Containers.
security	Security tooling (vasic-digital/Security).
helix_agent	AI-powered ensemble LLM service (HelixDevelopment/HelixAgent).
llms_verifier	Enterprise LLM verification platform — single source of truth for model metadata.
doc_processor	Document processing module (HelixDevelopment/DocProcessor).
llm_orchestrator	LLM orchestration module.
llm_provider	LLM provider abstraction module.
vision_engine	Vision / image-analysis engine.
debate_orchestrator	Multi-agent debate orchestration.
helix_specifier	Specification-handling module.
helix_memory	Memory subsystem.
helix_llm	LLM helper module.
rag	Retrieval-Augmented Generation module.
vector_db	Vector database abstraction.
embeddings	Embeddings generation/handling.
memory	Generic memory store.
agentic	Agentic-workflow primitives.
auth	Authentication module.
auto_temp	Auto-temperature tuning for LLM calls.
background_tasks	Background task execution (digital.vasic.background).
benchmark	Benchmarking utilities.
cache	Caching abstraction.
claritas	digital.vasic.claritas (clarity/analysis utility).
concurrency	Concurrency primitives (digital.vasic.concurrency).
config	Configuration module (digital.vasic.config).
conversation	

Submodule	Role (one-line)
	Conversation handling (digital.vasic.conversation).
database	Database abstraction (digital.vasic.database).
document	Document model (digital.vasic.document).
event_bus	Event bus (digital.vasic.eventbus).
filesystem	Filesystem abstraction (digital.vasic.filesystem).
formatters	Output/text formatters.
gandalf_solutions	GandalfSolutions utility module.
hyper_tune	Hyperparameter tuning.
i_llm	I-LLM interface module.
i18n	Internationalization (digital.vasic.i18n).
lazy	Lazy-evaluation utilities.
leak_hub	Leak/secret detection hub.
llm_ops	LLM operations module.
mcp_module	Model Context Protocol module.
messaging	Messaging module.
middleware	Middleware (digital.vasic.middleware).
models	Models module.
normalize	Normalization utilities (digital.vasic.normalize).
observability	Observability (digital.vasic.observability).
optimization	Optimization utilities.
ouroborous	Self-referential/loop orchestration module.
panoptic	Panoptic monitoring/observation module.
planning	Planning module.
plinius_common	PliniusCommon shared utilities.
plugins	Plugin framework.
rate_limiter	Rate limiting (digital.vasic.ratelimiter).
recovery	Application-level fault tolerance for Go.
red_team	Red-team / adversarial testing (digital.vasic.redteam).
self_improve	Self-improvement module.
skill_registry	Skill registry module.
storage	Storage abstraction (digital.vasic.storage).
streaming	Streaming module.
tool_schema	Tool schema definitions (digital.vasic.toolschema).
toon	digital.vasic.toon utility module.
veritas	Veritas verification/truth module.
watcher	File/resource watcher (digital.vasic.watcher).

Third-party upstream mirrors live under `dependencies/` (LLama_CPP, Ollama, HuggingFace_Hub). Reference agents/resources live under `cli_agents/` (50) and `cli_agents_resources/`.

4. Technology stack

Verified from `helix_code/go.mod` (inner module is the real dependency owner; cross-checked against CLAUDE.md §3.1). Actual versions found:

- **Language:** Go 1.26 (inner), Go 1.25.2 (root meta module).
- **HTTP / API / RPC:** gin-gonic/gin v1.11.0, gorilla/websocket v1.5.3, google.golang.org/grpc v1.80.0, grpc-ecosystem/grpc-gateway/v2 v2.28.0 (indirect).
- **Persistence:** jackc/pgx/v5 v5.7.6 + lib/pq v1.10.9 (PostgreSQL); redis/go-redis/v9 v9.17.2 (Redis); bradfitz/gomemcache; hashicorp/golang-lru/v2 v2.0.7.
- **AuthN/Z:** golang-jwt/jwt/v4 v4.5.2, golang.org/x/crypto v0.49.0 (bcrypt/argon2), golang.org/x/oauth2 v0.36.0.
- **Config / CLI:** spf13/viper v1.21.0, spf13/cobra v1.8.0, spf13/pflag v1.0.10, fsnotify/fsnotify v1.9.0.
- **LLM / cloud:** aws-sdk-go-v2 (v1.32.7) incl. service/bedrockruntime v1.23.1, aws/smithy-go v1.22.1; Azure azcore v1.16.0 + azidentity v1.8.0; getzep/zep-go/v3 v3.10.0; smacker/go-tree-sitter (tree-sitter parsing).
- **UI:** fyne.io/fyne/v2 v2.7.0 (desktop GUI); rivo/tview v0.42.0 + gdamore/tcell/v2 v2.8.1 (terminal UI); chromedp/chromedp v0.15.1 + chromedp/cdproto (headless browser).
- **LSP:** go.lsp.dev/protocol v0.12.0, go.lsp.dev/jsonrpc2 v0.10.0, go.lsp.dev/uri v0.3.0.
- **i18n / NLP:** nicksnyder/go-i18n/v2 v2.5.1, jdkato/prose/v2 v2.0.0.
- **Observability:** OpenTelemetry otel v1.43.0 family (OTLP gRPC/HTTP exporters, stdout exporters, SDK, metric, trace); go.uber.org/zap v1.28.0.
- **Testing:** stretchr/testify v1.11.1.
- **Own-org (via replace):** digital.vasic.containers, digital.vasic.helixqa, digital.vasic.debate, digital.vasic.helixspecifier, digital.vasic.lazy (and others wired through replace to ../submodules/).

Versions found in `go.mod` match the CLAUDE.md §3.1 stated versions where both list a number (Gin v1.11.0, gorilla/websocket v1.5.3, gRPC v1.80.0, JWT v4.5.2, Viper v1.21.0, Cobra v1.8.0, pflag v1.0.10, fsnotify v1.9.0, Fyne v2.7.0, testify v1.11.1).

5. Build & test entry points

Two Makefiles. The **root** Makefile runs governance gates only; the **inner** `helix_code/Makefile` does real builds/tests.

5.1 Root governance Makefile (verified targets)

bluff-detector, no-silent-skips (+ -warn), demo-all (+ -warn), demo-one, ci-validate-all, verify-foundation, verify-governance-cascade, verify-llmsverifier-pin-parity, scan-all, scan-gosec, scan-secrets (+ -root), scan-snyk, scan-sonarqube, scan-trivy.

5.2 Inner helix_code/Makefile (verified targets)

- **Build:** all, build, verify-compile, clean, clean-all, dev, dev-setup, prod, release, deps-scan, setup-deps, pgo-refresh.
- **Quality:** fmt, lint.
- **Test:** test, test-all, test-complete, test-coverage, coverage-full, test-benchmark, test-full, test-unit-full, test-integration-full, test-e2e-full, test-security-full, test-load-full, test-docs.
- **Test infra:** test-infra-up, test-infra-status, test-infra-down.
- **Verifier:** test-verifier-unit, test-verifier-integration, test-verifier-challenges, test-verifier-capability, test-verifier-hardcode.
- **Stress/chaos:** stress-chaos, stress-chaos-infra, stress-chaos-meta.
- **HelixQA:** helixqa-build, helixqa-test, helixqa-challenge, helixqa-bump-submodules.
- **Platforms:** desktop (+ -all/-linux/-macos/-windows/-nogui), mobile (+ -init/-ios/-android), aurora-os, harmony-os, aurora-harmony, windows.
- **Packaging:** installers, deb, rpm, homebrew, assets, logo-assets, logo-deps, docs, manual-html, sync-manual.
- **Containers (inner Makefile):** container-build, container-builder-image, container-clean, container-dev-up, container-dev-down, container-lint, container-release, container-shell, container-test, container-test-full. NOTE CLAUDE.md §3.4 (note 2026-05-29) states make container-* targets do not exist; the live inner Makefile DOES define them. Discrepancy flagged for reconciliation — per project policy containerized workflows go through the ./helix facade / containers submodule, but the targets are present in the file.
- **Security scans:** security-scan (+ -all/-gosec/-grype/-kics/-semgrep/-snyk/-sonarqube/-trivy), secrets-scan, scan-start-sonar, scan-stop.

5.3 ./helix container facade

A repo-root bash script (helix) providing a unified interface to the containerized stack (start/stop/status/logs/restart/shell). NOTE the script's own header says "Docker Facade" and its check_docker() invokes docker; CLAUDE.md states the host uses podman, not docker. The script as written checks for docker on PATH — discrepancy flagged (the documented podman workflow and the script's docker invocation are not aligned in the inspected header).

6. Governance / constitution cascade

The **constitution submodule** at constitution/ is the canonical root of governance (per CONST-059 / §11.4.35). Verified present: constitution/Constitution.md, constitution/CLAUDE.md, constitution/AGENTS.md (each with .html/.pdf/.docx sibling exports).

The consumer-side repo-root files — CLAUDE.md, AGENTS.md, CONSTITUTION.md, plus sibling agent manuals CRUSH.md, QWEN.md — are **extensions** that inherit from the canonical root. CLAUDE.md opens with @constitution/CLAUDE.md / an ## INHERITED FROM constitution/CLAUDE.md pointer. Project-specific rules live consumer-side; universal rules live in the constitution submodule. Cascade is verified via ./scripts/verify-governance-cascade.sh (root governance gate verify-governance-cascade).

7. Known-incomplete / honest gaps

This section records areas the live tree shows as in-progress or inconsistent. It does NOT claim any feature works; it records observable structural facts.

1. **CLAUDE.md §3.2 vs live submodule layout.** §3.2 documents a flat repo-root layout for several owned submodules; the live tree uses submodules/<name>/. CLAUDE.md §3.2 is stale relative to the tree and should be reconciled.
2. **.gitmodules section-name vs path mismatch.** Several entries have section names like [submodule "dependencies/HelixDevelopment/llm_orchestrator"] but path = submodules/llm_orchestrator. The naming is inconsistent (legacy section labels not updated after the relayout). Paths are authoritative.
3. **submodules/* directories not all in .gitmodules.** 63 directories exist under submodules/, but only a subset have formal .gitmodules entries (e.g. containers, helix_qa, challenges, security, llms_verifier, helix_agent, doc_processor, llm_orchestrator, llm_provider, vision_engine). The remainder are present in the working tree without verified .gitmodules wiring in the file inspected — the submodule registry and the on-disk tree are not fully in sync.
4. **Inner container-* Makefile targets vs CLAUDE.md §3.4.** CLAUDE.md says these targets were removed and do not exist; the inner Makefile still defines them. One of the two is out of date.
5. **./helix facade uses docker, policy mandates podman.** The script header and check_docker() invoke docker; project rules (Rule 4 / §11.4.76) mandate podman via the orchestrator. The facade script is not aligned with the stated podman workflow.
6. **Applications directory naming.** Live dirs use snake_case (aurora_os, harmony_os, terminal_ui); CLAUDE.md §3.2.1 lists hyphenated names. CLAUDE.md is stale here (the live tree is the CONST-052-compliant form).
7. **Large volume of root-level status/report .md files.** The repo root holds many *_COMPLETION_*, *_REPORT.md, PHASE_* documents. These are historical narrative artifacts; they are NOT authoritative for current architecture (this document is). Their accuracy is not verified here.
8. **Duplicate-looking internal packages.** internal/i18n_wiring and internal/i18nwiring both exist; internal/provider and internal/

providers both exist. Whether this is intentional layering or in-progress consolidation was not determined from directory listing alone.

**Sources verified 2026-06-04: /Users/milosvasic/
Projects/HelixCode/.gitmodules, /Users/milosvasic/
Projects/HelixCode/go.mod, /Users/milosvasic/
Projects/HelixCode/helix_code/go.mod, /Users/
milosvasic/Projects/HelixCode/Makefile, /Users/
milosvasic/Projects/HelixCode/helix_code/Makefile, /
Users/milosvasic/Projects/HelixCode/helix (facade
script header), ls of repo root, ls of submodules/, ls of
helix_code/{cmd,internal,applications,tests}, ls of
cli_agents/ + submodules/ (counts), README.md
first headings of submodules/* for the catalogue, ls of
constitution/.**